

# A Computer-Assisted Seven-Point Candidate Refinement for Simple Zeros of the Riemann Zeta Function

Anonymous submission  
AI-assisted research draft

Peer-review draft, August 12, 2026

**Release status.** This manuscript is an unreviewed, computer-assisted candidate refinement. It does not prove the Riemann hypothesis. The finite seven-point inequality has been verified at two Arb precision settings. A pinned Lean 4 companion checks the new finite-dimensional stability, spectral-energy, and exact assembly lemmas without `sorry`. It also checks an abstract defect-preserving Theorem D source inequality, with the remaining source error proved to be  $o(N)$ , as well as the finite consecutive-window count, arbitrary principal-block spectral pinching, abstract 267-offset aggregation, and a conditional 267-point block bridge. It also enumerates the actual simple critical-line Gram columns by increasing ordinate and proves that this reindexing preserves the full spectral defect. Each full consecutive 267-point principal block and its normalized adjacent gaps are now concrete Lean objects, with positivity and telescoping spans checked. The 267 concrete shifted partitions, their endpoint-remainder fibers, global pinching, and insertion into the exact finite simple-zero count are also checked. It does not yet replay the Arb certificate or close endpoint deletion, the summably uniform Gram-to-kernel comparison, and the asymptotic top-level specialization interfaces. The analytic trace and tail estimates are imported from Claude’s Theorem D [1]. No part of the proposed improvement has yet received independent expert review.

## Abstract

Let  $N(T, 2T)$  count the nontrivial zeros of the Riemann zeta function with ordinates in  $(T, 2T]$ , with multiplicity, and let  $N_0^s(T, 2T)$  count the simple zeros on the critical line in the same interval. We retain a convex spectral defect in the rank–trace argument underlying Claude’s 2026 Montgomery–Taylor bound and connect that defect to short-range correlations of consecutive critical-line zeros. The only new computer-assisted input is the global inequality

$$\mathcal{F}_6(g_1, \dots, g_6) \geq C_* := \frac{38262312113}{10^{13}} \quad (g_1, \dots, g_6 \geq 0),$$

for an explicit seven-point functional built from the normalized Montgomery–Taylor overlap kernel. An exhaustive interval verifier uses Arb enclosures, certified convex tangent bounds, and two rational strong-convexity basins; a higher-precision replay has identical logical counts. A Lean companion formalizes the new matrix and exact-arithmetic layers, the finite window/pinching/267-offset core, and an abstract defect-preserving Theorem D source inequality. It also formalizes the increasing-ordinate reindexing of the concrete simple-zero Gram matrix and its exact defect invariance, and specializes the local block bridge to every concrete consecutive 267-point principal block. It also realizes all 267 shifted partitions and endpoint fibers on the actual ordered index, proves their global pinching bound, and inserts the resulting finite defect bound into the exact simple-zero count. It leaves the local certificate replay, endpoint deletion, summably uniform

Gram-to-kernel error, and asymptotic top-level specialization as explicit open obligations. Subject to those stated verification obligations, the candidate bound is

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.6730254768378743181159739993 \dots$$

## 1 Scope and statement of the result

Write

$$N := N(T, 2T), \quad S := N_0^s(T, 2T),$$

and let

$$H_{\text{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457343797908 \dots \quad (1)$$

The quantity  $H_{\text{MT}}$  is the optimized constant in Theorem D of Claude [1]. That theorem is accompanied by a Lean 4 artifact for its stated results [2]. We use its analytic framework as a cited input. The accompanying **ZetaSeven** package is a pinned extension of that Lean development: it verifies the new finite-dimensional lemmas, the finite window/pinching/267-offset core, the defect-preserving abstract Theorem D source inequality, concrete shifted partitions with endpoint fibers, their finite simple-count assembly, and exact algebra, but does not yet prove the local seven-point claim or the strengthened end-to-end top-level statement.

The exact local constant certified in this work is

$$C_* := \frac{38262312113}{1000000000000} = 0.0038262312113. \quad (2)$$

**Proposition 1.1** (Relative candidate deduction). *Assume the analytic trace, endpoint-tail, inertia, and uniform overlap conclusions used in the proof of Theorem D of [1]. Then*

$$\begin{aligned} \liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} &\geq \frac{267H_{\text{MT}} - 266/500}{267 - 261C_*} \\ &= \frac{2670000000000000H_{\text{MT}} - 532000000000}{2660013536538507} \\ &= 0.67302547683787431811597399932145 \dots \end{aligned} \quad (3)$$

*Equivalently, the lower proportion is 67.3025476837874318115...%.*

The adjective “relative” is a disclosure about verification scope, not an additional number-theoretic hypothesis such as RH. If the cited analytic inputs are accepted, the deduction below is unconditional. This manuscript itself, however, is not yet a peer-reviewed validation of those inputs or of their interface with the new defect term.

The proof has three new layers. Section 3 preserves a nonnegative spectral defect in the matrix inequality. Sections 4–5 lower-bound that defect using the Montgomery–Taylor kernel and a rigorous seven-point computation. Sections 6–7 convert the local inequality to the exact constant in Proposition 1.1.

## 2 The imported analytic interface

We state precisely the portion of [1] used by the refinement. Put

$$\ell := \log \frac{T}{2\pi}, \quad L := \ell, \quad I := (T, 2T], \quad I' := (T - T^{1/2}, 2T + T^{1/2}).$$

For the optimized test family of [1, Section 7.1], let  $\hat{A}$  be the normalized finite compression of Weil's Hermitian form. Partition the zeros in  $I'$  into simple critical-line zeros  $\mathcal{S}_1$ , distinct multiple critical-line zeros  $\mathcal{S}_2$ , and unordered off-line pairs  $\mathcal{P}$ . Write

$$s_1 := \#\mathcal{S}_1, \quad s_2 := \#\mathcal{S}_2, \quad p := \#\mathcal{P}.$$

Counting multiplicity gives

$$N(I') \geq s_1 + 2s_2 + 2p. \quad (4)$$

For  $\rho \in \mathcal{S}_1$ , let  $v_\rho$  be the normalized Gabor atom of [1, Sections 2 and 4], and let  $V$  have these atoms as columns. Then

$$P_1 = VV^*, \quad M = V^*V,$$

the column norms are at most one, and for  $Q' := \hat{A} - P_1$  the inertia decomposition gives

$$n_+(Q') \leq s_2 + p. \quad (5)$$

The trace and tail analysis in [1, Sections 4–7] yields the baseline constant (1). In the form needed below, once the matrix inequality of Section 3 is applied, the same estimates give

$$S \geq H_{\text{MT}}N + \mathcal{D}(M^\circ) - o(N), \quad (6)$$

where  $M^\circ$  is the Gram matrix of the retained central simple zeros and  $\mathcal{D}$  is defined in (7) below. We derive the added term  $\mathcal{D}(M^\circ)$  in Corollary 3.2; all analytic error estimates used to pass to (6) remain those of the cited paper.

The second imported interface is the full-grid identity of [1, Lemma 2.2] and its endpoint truncation. It gives the uniform kernel limit stated in Lemma 4.1. The normalization and closed form of that kernel are rederived in Section 4, since they are also the input to the interval verifier.

### 3 A stability refinement of the rank–trace inequality

Define the convex, nonnegative function

$$\Psi(t) := \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2, \end{cases} \quad \mathcal{D}(M) := \text{tr } \Psi(M) \quad (7)$$

for positive semidefinite  $M$ .

**Lemma 3.1** (Stability-enhanced rank–trace inequality). *Let  $V$  be a  $d \times r$  matrix whose columns have norm at most one. Put  $P = VV^* \succeq 0$  and  $M = V^*V$ , and let  $Q$  be Hermitian with  $n_+(Q) \leq b$ . Then*

$$\|P + Q\|_{\text{F}}^2 \geq 4 \text{tr}(P + Q) - 3r - 4b + \mathcal{D}(M). \quad (8)$$

*Proof.* Write  $Q = Q_+ - Q_-$  with  $Q_\pm \succeq 0$ ,  $Q_+Q_- = 0$ , and  $\text{rank } Q_+ \leq b$ . Since  $P, Q_+ \succeq 0$ ,

$$\|P + Q\|_{\text{F}}^2 \geq \|P - Q_-\|_{\text{F}}^2 + \|Q_+\|_{\text{F}}^2.$$

For every  $q \geq 0$ ,  $q^2 \geq 4q - 4$ , hence

$$\|Q_+\|_{\text{F}}^2 \geq 4 \text{tr } Q_+ - 4b. \quad (9)$$

Let  $p_1 \geq \dots \geq p_r \geq 0$  be the eigenvalues of  $M$ , padding the nonzero spectrum of  $P$  by zeros, and let  $n_1 \geq n_2 \geq \dots \geq 0$  be the eigenvalues of  $Q_-$ . Von Neumann's trace inequality gives  $\text{tr}(PQ_-) \leq \sum_{i \leq r} p_i n_i$ . Therefore

$$\|P - Q_-\|_{\text{F}}^2 + 4 \text{tr} Q_- \geq \sum_{i \leq r} ((p_i - n_i)^2 + 4n_i).$$

For  $p \geq 0$ ,

$$\min_{n \geq 0} ((p - n)^2 + 4n) = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + \Psi(p). \quad (10)$$

It follows that

$$\|P - Q_-\|_{\text{F}}^2 \geq 2 \text{tr} P - r + \mathcal{D}(M) - 4 \text{tr} Q_-. \quad (11)$$

Adding (9) and (11) gives

$$\|P + Q\|_{\text{F}}^2 \geq 4 \text{tr}(P + Q) - 2 \text{tr} P - r - 4b + \mathcal{D}(M).$$

Finally  $\text{tr} P$  is the sum of the squared column norms of  $V$ , so  $\text{tr} P \leq r$ .  $\square$

**Corollary 3.2** (Global defect inequality). *With the notation of Section 2,*

$$s_1 \geq 4 \text{tr} \hat{A} - \|\hat{A}\|_{\text{F}}^2 - 2N(I') + \mathcal{D}(M). \quad (12)$$

*After the endpoint truncation and analytic estimates of [1],*

$$S \geq H_{\text{MT}} N + \mathcal{D}(M^\circ) - o(N). \quad (13)$$

*Proof.* Apply Lemma 3.1 to  $P_1 + Q' = \hat{A}$ . Equations (4) and (5) imply

$$\begin{aligned} \|\hat{A}\|_{\text{F}}^2 &\geq 4 \text{tr} \hat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \\ &\geq 4 \text{tr} \hat{A} - s_1 - 2N(I') + \mathcal{D}(M), \end{aligned}$$

which is (12).

Now use the same comparison of  $\hat{A}$  with the prime-side matrix, the same Frobenius and trace asymptotics, and the same tail estimates as in the proof of Theorem D of [1]. Pinching  $M$  into the retained central block and its complement gives  $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$ : pinching is an average of unitary conjugations,  $X \mapsto \text{tr} \Psi(X)$  is convex and unitarily invariant, and  $\Psi \geq 0$ . The remaining main term is  $H_{\text{MT}} N$ , yielding (13).  $\square$

## 4 The Montgomery–Taylor overlap kernel

Let  $h := 2\pi/L$  and normalize ordinates by

$$x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

**Lemma 4.1** (Uniform local overlap). *Fix  $R_0 > 0$ . After deleting simple zeros in normalized endpoint strips of width  $L^2$ , the number deleted is  $o(N)$ , and uniformly for retained simple zeros with  $|x_\rho - x_{\rho'}| \leq R_0$ ,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1), \quad (14)$$

where

$$k(x) := \frac{K(x)}{K(0)}, \quad (15)$$

$$\begin{aligned} K(x) &:= \int_{-1/2}^{1/2} \cos(\sqrt{2}t) \cos(2\pi xt) dt \\ &= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \\ K(0) &= \sqrt{2} \sin(1/\sqrt{2}). \end{aligned} \quad (16)$$

The apparent singularities in (16) are removable.

*Proof.* The full-grid Gabor identity in [1, Lemma 2.2] expresses the normalized inner product through the Fourier transform of the squared window. For the optimized window, substitution  $u = Lt$  gives

$$\frac{\int_{-1/2}^{1/2} \phi(Lt)^2 e^{-2\pi ixt} dt}{\int_{-1/2}^{1/2} \phi(Lt)^2 dt} \rightarrow \frac{K(x)}{K(0)}.$$

Indeed, the fixed-width endpoint ramp has rescaled length  $O(L^{-1})$ , while  $\phi(Lt)^2$  tends in  $L^1$  to  $\cos(\sqrt{2}t)\mathbf{1}_{[-1/2, 1/2]}(t)$ . The convergence is uniform for  $x$  in compact sets. The endpoint-tail estimate used in [1, Section 5.3] is uniform for bounded separations. A normalized strip of width  $L^2$  has physical width  $O(L)$  and contains  $O(L^2) = o(N)$  zeros by the standard local zero count. This proves (14). The elementary integral gives (16) and its value at zero.  $\square$

Set

$$w(x) := k(x)^2. \quad (17)$$

The verifier uses the equivalent entire expression

$$K(x) = \frac{1}{2} \left( \operatorname{sinc}(\pi x - 1/\sqrt{2}) + \operatorname{sinc}(\pi x + 1/\sqrt{2}) \right), \quad (18)$$

where  $\operatorname{sinc}(z) = \sin z/z$  with its continuous value at zero.

## 5 The certified seven-point inequality

For six nonnegative consecutive gaps define

$$\mathcal{F}_6(g_1, \dots, g_6) := \frac{1}{3000} \sum_{i=1}^6 g_i + \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \dots + g_{i+r-1}). \quad (19)$$

The second term contains all 21 pair separations among seven ordered points.

**Proposition 5.1** (Certified local inequality). *For all  $g_1, \dots, g_6 \geq 0$ ,*

$$\mathcal{F}_6(g_1, \dots, g_6) \geq C_* = \frac{38262312113}{10^{13}}. \quad (20)$$

*Computer-assisted verification.* The accompanying verifier uses `python-flint==0.8.0` and Arb interval arithmetic [3]. Its grid mesh is  $1/4000$ . On every closed cell it evaluates (18), forms a rigorous lower bound for  $w$ , and widens conversion to IEEE-754 binary64 downward. A separate interval table lower-bounds  $w''$ . All later range queries use minima of those cell bounds.

The exact pressure cutoff is

$$q = \lceil C_* \cdot 4000 \cdot 3000 \rceil = 45915.$$

Thus  $\sum_i g_i/3000$  proves the target whenever the sum of lower cell indices is at least  $q$ . Before forming six-dimensional boxes, the verifier also uses the one-variable contribution

$$U(g) := \frac{g}{3000} + \frac{1}{3}w(g).$$

The surviving cells form the three exact components

$$[3807, 4780], \quad [7218, 9373], \quad [10560, 44945].$$

Pressure filtering leaves 435 initial products of components.

For each box, every consecutive partial sum in (19) receives a range-minimum lower bound. If this direct interval bound is insufficient, the code lower-bounds the Hessian throughout the box using the  $w''$  table. A binary64 LDL factorization is only a cheap filter; positive definiteness is rechecked with Arb. On a certified convex box, evaluation at its rational midpoint gives the rigorous tangent bound

$$\mathcal{F}_6(x) \geq \mathcal{F}_6(c) - \sum_{i=1}^6 |\partial_i \mathcal{F}_6(c)| r_i.$$

Two neighborhoods of the numerical minimizers require a sharper local argument. The first has exact rational center represented by

$$c = (1.04608035577, 1.98913202062, 1.98641493611, \\ 1.04160329372, 1.97702352234, 1.04500209462). \quad (21)$$

and the second has reversed coordinates. Each coordinate radius is exactly  $1/64$ . Interval lower bounds for  $w''$ , followed by an Arb LDL check, prove throughout both basins that

$$\nabla^2 \mathcal{F}_6 \succeq \frac{3}{16} I. \quad (22)$$

At 256-bit basin precision, strong convexity gives

$$\inf_{\text{basin}} \mathcal{F}_6 \geq \mathcal{F}_6(c) - \frac{\|\nabla \mathcal{F}_6(c)\|^2}{2(3/16)} \\ \geq 0.00382623121130447424827371713006270699383823035, \quad (23)$$

and the same lower endpoint for the reversed basin. This clears the exact target by more than  $4.47 \times 10^{-15}$ . A search box is basin-pruned only when all of its closed grid cells lie inside a certified basin.

All other boxes are bisected along a widest coordinate. Reaching an unresolved terminal cell raises an exception. The canonical run has the following deterministic counts.

| Field                     | Canonical value |
|---------------------------|-----------------|
| Initial boxes             | 435             |
| Visited nodes             | 822,433         |
| Pruned leaves             | 411,434         |
| Splits                    | 410,999         |
| Maximum depth             | 39              |
| Interval prunes           | 285,258         |
| Pressure prunes           | 2,872           |
| Tangent prunes            | 122,329         |
| Strong-convexity prunes   | 975             |
| Unresolved terminal cells | 0               |

The code checks both binary-forest identities

$$822433 = 411434 + 410999, \quad 411434 = 410999 + 435.$$

The canonical kernel-table SHA-256 is

5a1f95f754a83ba05f37692d4bda69bf3fa5d3752af902826bfc4cffd31428b9,  
and the second-derivative-table SHA-256 is  
318ee79f619cc75c6f7a30424a181764865910693f33a3d95469ebcae0ce42ba.

A full rerun with table precision increased from 128 to 160 bits and basin precision from 256 to 320 bits produces the same initial boxes, node counts, pruning counts, and depth. Its independently regenerated second-derivative table has SHA-256

fd634c87441380105aad52f069f735108d0dabcd7d5d0a993b6abc52ee12b055.

This is a cross-precision stress test, not an independent implementation: it uses the same source and Arb/FLINT library. The exact expected reports are included in the artifact and are compared field by field on every release run. These facts complete the finite verification of Proposition 5.1.  $\square$

**Remark 5.2.** *The center (21) was located by ordinary floating-point optimization, but the optimizer output is not trusted. Only the displayed terminating decimals, interpreted as exact rationals, and the interval proof of (22)–(23) enter the certificate.*

## 6 From seven-point windows to block defect

Let  $y_1 < \dots < y_m$  and set

$$E_m := 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i). \quad (24)$$

**Lemma 6.1** (Block energy). *For every  $m \geq 7$ ,*

$$E_m + \frac{1}{500}(y_m - y_1) \geq C_*(m - 6). \quad (25)$$

*Proof.* Write  $g_i = y_{i+1} - y_i$  and sum (20) over the  $m - 6$  consecutive seven-point windows. A pair spanning  $r \leq 6$  gaps occurs in at most  $7 - r$  windows and has coefficient  $2/(7 - r)$  in each, so its total coefficient is at most 2. Pairs spanning more than six gaps do not appear and contribute nonnegatively to  $E_m$ . Each single gap occurs in at most six windows, so its total linear coefficient is at most  $6/3000 = 1/500$ .  $\square$

**Lemma 6.2** (Block defect). *If  $G \succeq 0$  is Hermitian, then*

$$\mathrm{tr} \Psi(G) \geq \min \left\{ 1, 2 \sum_{i < j} |G_{ij}|^2 \right\}. \quad (26)$$

*Proof.* If every eigenvalue of  $G$  is at most 2, then  $\Psi(G) = (G - I)^2$  and

$$\mathrm{tr} \Psi(G) = \mathrm{tr}(G - I)^2 \geq 2 \sum_{i < j} |G_{ij}|^2.$$

If some eigenvalue  $\lambda > 2$ , then  $\Psi(\lambda) = 2\lambda - 3 > 1$ , while all other contributions are nonnegative.  $\square$

Choose

$$m := 267, \quad A_0 := C_*(m - 6) = 0.9986463461493 < 1. \quad (27)$$

For comparison,

$$C_*(268 - 6) = 1.0024725773606 > 1. \quad (28)$$

Thus  $m = 267$  is the last block size before the unit cap in Lemma 6.2 activates; direct substitution also shows that it maximizes the resulting bound among integer block sizes  $m \geq 7$ .

Let  $B$  be a block of 267 consecutive retained central simple zeros, and let  $G_B$  be its Gram matrix. If  $\mathrm{span}(B)/500 \geq A_0$ , the next inequality is immediate. Otherwise  $\mathrm{span}(B) < 500$ , so Lemma 4.1 applies uniformly to every pair in the fixed-size block and

$$2 \sum_{i < j} |(G_B)_{ij}|^2 = E_m + o(1).$$

Lemmas 6.1 and 6.2 give, uniformly in  $B$ ,

$$\mathcal{D}(G_B) + \frac{1}{500} \mathrm{span}(B) \geq A_0 - o(1). \quad (29)$$

## 7 Shifted block pinching and final arithmetic

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalized ordinates of the retained central simple zeros. Endpoint deletion gives

$$S^\circ = S - o(N). \quad (30)$$

For each of the  $m = 267$  offsets, partition the ordered points into full consecutive blocks of size  $m$ , leaving only bounded endpoint remainders. Pinching  $M^\circ$  to the principal blocks and the remainder gives

$$\mathcal{D}(M^\circ) \geq \sum_B \mathcal{D}(G_B). \quad (31)$$

For each offset, sum (29) over its full blocks and use (31); then average over all  $m$  offsets. The average number of full blocks is  $S^\circ/m + O(1)$ . Each adjacent gap belongs to a block span for at most  $m - 1$  offsets. Moreover

$$x_{S^\circ} - x_1 \leq \frac{LT}{2\pi} = N + o(N) \quad (32)$$



by the Riemann–von Mangoldt formula. The blockwise error in (29) is uniform, so its total is  $o(N)$ . Consequently

$$\mathcal{D}(M^\circ) \geq \frac{A_0}{m}S - \frac{m-1}{500m}N - o(N) = \frac{261C_*}{267}S - \frac{266}{133500}N - o(N). \quad (33)$$

Substitute (33) into (13):

$$S \geq H_{\text{MT}}N + \frac{261C_*}{267}S - \frac{266}{133500}N - o(N).$$

Hence

$$\left(1 - \frac{261C_*}{267}\right)S \geq \left(H_{\text{MT}} - \frac{266}{133500}\right)N - o(N). \quad (34)$$

Divide by  $N$ , let  $T \rightarrow \infty$ , and use (2). This gives (3). Multiplying numerator and denominator by  $267 \cdot 10^{13}$  gives the exact reduced expression in Proposition 1.1; the common integer gcd is one. This proves the relative candidate deduction, subject to the stated imported interfaces and open verification obligations.

## 8 Position relative to prior work

Montgomery’s pair-correlation argument proved, under RH, that at least two thirds of the zeros are simple [4]; the optimized Montgomery–Taylor window gives the constant (1) [5]. Chirre, Gonçalves, and de Laat later obtained stronger conditional simple-zero estimates by semidefinite programming and positivity outside the bandwidth-one interval [6]. Those results operate in a different conditional regime and are not numerically comparable to Proposition 1.1 as an unconditional critical-line statement.

Recent work of Goldston and collaborators clarified how pair correlation can control zeros that are simultaneously simple and on the critical line, under the pair-correlation conjecture or narrow-box hypotheses [7, 8, 9]. Claude’s paper [1] then supplied the unconditional finite-dimensional inertia framework and the baseline  $H_{\text{MT}}$  used here. The present refinement does not extend Fourier support or add a new prime-side estimate. It extracts extra information from the short-range Gram matrix by preserving the spectral defect  $\mathcal{D}$  and certifying a seven-point local inequality.

The software lineage is also relevant. A public AI-generated artifact first implemented the stability approach at local target 0.0038 [10]; a clean-room release certified 0.00382 [11]. The current artifact certifies (2) and adds the explicit strong-convexity basins and precision replay. These repositories are research artifacts, not substitutes for peer-reviewed priority analysis. A search of the cited 2025–2026 primary literature found no identical stability-defect/seven-point construction, but that search cannot establish priority.

## 9 Reproducibility, trust base, and limitations

Proposition 5.1 is the only computer-assisted proposition in the deduction. The accompanying source archive contains the verifier, exact JSON reports, unit tests, a hash-locked dependency file, and detailed audit and reproduction notes. The canonical commands are

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --require-hashes -r requirements.lock
```

```
PYTHONPATH=src python -m unittest discover -s tests -v
PYTHONPATH=src python scripts/verify_release.py
PYTHONPATH=src python scripts/verify_cross_precision.py
```

The two verifier scripts agree with their expected reports when they end with the respective status strings

```
release_certificate=match
cross_precision_certificate=match.
```

The partial Lean companion uses Lean v4.33.0-rc2 and is pinned to the following upstream and mathlib commits:

```
upstream: 3635e74826a4c1fcede7d1cd2b6fa75e43a00510
mathlib: 51e6992efd06126df61a496bebf8f49482a4e129.
```

Its audit commands are

```
lake build ZetaSeven
lake env lean ZetaSeven/Audit.lean
```

The proved declarations cover the rank–trace surplus, the simple-zero Gram split and its inertia bound, the finite tail seam, the bridge from off-diagonal Gram energy to spectral defect, the exact consecutive-window count, arbitrary principal-block spectral pinching, abstract 267-offset aggregation, the conditional 267-point block bridge, the abstract defect-preserving Theorem D source inequality (including an explicit  $o(N)$  error package), the concrete shifted partitions and endpoint fibers, their insertion into the exact finite simple-zero count, and the exact rational and epsilon-form assembly. The proposition `SevenPointClaim` is intentionally defined but not asserted as a theorem. Consequently, a successful Lean build is evidence for these layers only; it is not an end-to-end formal proof of Theorem 1.1.

The finite trust base consists of CPython 3.12, IEEE-754 binary64 semantics, `python-flint` version 0.8.0, Arb/FLINT, the operating system and hardware, and the included source. The verifier rebuilds all transcendental tables; it does not trust cached tables, the numerical optimizer, or committed run logs. No second implementation or proof-assistant checker is yet available for the new local inequality. Lean reduces the trust base for the surrounding algebra but does not reduce the Arb certificate’s trust base until a proof-carrying certificate checker is completed.

The larger mathematical trust boundary is equally important. This work does not independently rebuild the analytic estimates of [1] or its Lean artifact. The new Lean theorem checks preservation of  $\mathcal{D}(M^\circ)$  through an abstract package of the pinned trace and tail hypotheses and packages the remaining source-side terms as  $o(N)$ . Reviewers should still audit the correspondence of those hypotheses with the upstream concrete parameter family, as well as endpoint deletion and the summably uniform Gram-to-kernel replacement in the remaining analytic bridge.

For peer review, the highest-value checks are:

1. verify Lemma 3.1 and the deduction of Corollary 3.2 against the exact notation of [1];
2. independently inspect the cell-covering, outward-rounding, Hessian, and basin-containment logic in the verifier;
3. reproduce both certificates on a clean machine and, preferably, a second architecture;
4. implement an independent interval or proof-assistant checker for Proposition 5.1;

5. check the seven-window multiplicities and shifted gap charges in Sections 6–7;
6. repeat the prior-art search before any public priority claim.

Finally, neither a positive proportion result nor the local statistics used here imply that all zeros lie on the critical line. The method remains far from a proof of the Riemann hypothesis, and the bandwidth-one framework has known structural ceilings discussed in [1].

## Acknowledgment and authorship placeholder

This anonymous line is intended for review circulation only. A submission must name the people who take responsibility for the mathematics, disclose the AI-assisted origin in accordance with the venue’s policy, and obtain their approval of the final text. The upstream software credits and license are recorded in the accompanying `PROVENANCE.md`.

## References

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